SJ Thanav Kumar

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Education

Shiv Nadar University, Chennai Aug 2024 – Present
B.Tech in Computer Science and Engineering (IoT), CGPA: 8.14

Vidya Mandir Senior Secondary School May 2016 – 2024
CBSE – Class XII: 477/500 (2024): 95.4% — *CBSE – Class X: 410/500 (2022): 82%*

Work Experience

Research Intern May 2026 – Jun 2026
NIT Trichy

- Devised four different deep learning model architectures (DETR-style point cloud decoder, UNet heatmap decoder, coordinate-conditioned MLP, and a ResNet-18 CNN variant) to reconstruct the 3D surface of the human cornea from multi-angle eye images.
- Built a Swin Transformer encoder with cross-view attention as the shared backbone, and generated a synthetic Blender dataset to train and validate the models.
- Produced clinical topography outputs (curvature, keratometry, and elevation maps) alongside a CSV of predicted XYZ coordinates for 3D surface rendering.

Projects

Quarry – Local Image and Video Converter | *Python, PySide6, Pillow, FFmpeg* Oct 2026

- Built a cross-platform desktop app that converts and compresses images (PNG, JPEG, WebP, AVIF, HEIC, and more) and video (MP4, MOV, MKV) offline, with quality and CRF sliders, H.264/H.265 encoding, video-to-GIF conversion, and animated GIF/WebP/AVIF support.
- Architected a GUI-independent engine around a `ConversionManager` with pluggable Pillow and FFmpeg backends, handling magic-byte format detection, safe temp-file writes, batch processing with per-file error isolation, and cancellation; video remuxes without re-encoding when codecs allow.

ClearCash – Financial Assistant for MSMEs | *SNUC 2026 Hacks* Feb 2026

- Designed a high-fidelity, institutional-grade financial intelligence dashboard for MSMEs featuring automated parsing, liquidity analysis, and predictive decision engines to convert raw banking data into financial strategy.
- Built a skeleton model within the hackathon timeframe, laying out the core architecture for the full ecosystem.

GAYME – Multi-Game Arcade | *Python, Pygame*

- Led and coordinated a team to build a mini-arcade of six games, including Super TicTacToe (a recursive/dynamic-programming nested tic-tac-toe variant), FlappyMan, DinoRun, SpaceRaider, and a 2–3 player Smash Bros-style co-op clone.
- Developed a multiplayer concept game to demonstrate real-time multiplayer capability within Pygame.

Blackjack Simulator | *Python, OpenCV – SNUC 2024 Internal Hackathon* 2024

- Built a card-recognition pipeline using a custom image dataset to identify player and dealer cards via camera input.
- Computed optimal play (hit/stand) in real time using probability-based decision logic, built to teach blackjack strategy using IRL tools.

Technical Skills, Language Skills, and Interests

Programming Languages: C, C#, C++, Python, Java, SQL
Web Technologies: HTML, CSS, JS
Operating Systems: Windows, macOS
Tools & Software: Git, Blender, Unity, Godot, FFmpeg
Libraries: OpenCV, Tkinter, Pygame, PySide6, Pillow
Languages: English, Tamil

Hackathons

SNUC 2024 Internal Hackathon ◇ SNUC 2026 Hacks